



# JAVA SNACKS – STRUCTURES

## (Saturday, May 2, 2026)

### Instructions:

1. **Avoid using Artificial Intelligence(AI) tools.**
2. **Submission must occur before class (8:30am – Monday, May 4, 2026).**
3. **Each question is to be in a different file.**

**Submission Guideline: It is safe to assume you have created a repo called JavaTasks(or its equivalent).**

**Create a new folder **structures** and save all of today's tasks inside it**

**Push to git.**

1. Write a method with the header: `public static int getPentagonalNumber(int n)` that returns the nth pentagonal number using the formula  $n(3n-1)/2$ . Write a test program that displays the first 100 pentagonal numbers, 10 per line, using the `%7d` format specifier.
2. Write a method with the header: `public static int sumDigits(long n)` that computes and returns the sum of the digits in the integer n. Write a test program that prompts the user to enter an integer and displays the sum of all its digits.
3. Write a method with the header: `public static int reverse(int number)` that returns the reversal of an integer (e.g., `reverse(456)` returns 654). Then write a method: `public static boolean isPalindrome(int number)` that uses `reverse` to return true if the number is a palindrome. Write a test program that prompts the user to enter an integer and reports whether it is a palindrome.
4. Write a method with the header: `public static void reverse(int number)` that displays an integer in reverse order (e.g., `reverse(3456)` displays 6543). Write a test program that prompts the user to enter an integer and displays its reversal.
5. Write a method with the header: `public static void displaySortedNumbers(double num1, double num2, double num3)` that displays three numbers in decreasing order. Write a test program that prompts the user to enter three numbers and invokes this method.
6. Write a method with the header: `public static void displayPattern(int n)` that displays a staircase pattern where row 1 shows '1', row 2 shows '1 2', up to row n which shows 'n n-1 ... 2 1'. Write a test program that prompts the user for n and calls the method.
7. Write a method with the header: `public static double futureInvestmentValue(double investmentAmount, double monthlyInterestRate, int years)` that returns the future investment value. Write a test program that prompts the user for investment amount and annual interest rate, and displays a table of future values for years 1 through 30.
8. Write a class containing the two methods: `public static double mileToKilometer(double mile)` and `public static double kilometerToMile(double kilometer)`. Write a test program that converts 2 miles to kilometers and 2 kilometers to miles, and displays a conversion table from 1 to 10 for each direction.



9. Write a method: `public static double computeTax(int status, double taxableIncome)` that computes tax based on filing status. Use this method in a program that prints a tax table for taxable income from \$50,000 to \$70,000 at \$100 intervals, showing taxes for all four filing statuses.
10. Write a method with the header: `public static int numberOfDaysInAYear(int year)` that returns the number of days in a given year (365 or 366). Write a test program that displays the number of days in years from 2014 to 2034.
11. Write a method with the header: `public static void printMatrix(int n)` that displays an n-by-n matrix of randomly generated 0s and 1s. Write a test program that prompts the user to enter n and calls the method.
12. Write a method: `public static boolean isValidPassword(String password)` that returns true if the password has at least 10 characters, consists only of letters and digits, and contains at least 3 digits. Write a test program that prompts the user to enter a password and displays: Valid Password or Invalid Password.
13. Write two methods: `public static boolean isValid(double side1, double side2, double side3)` that returns true if three sides form a valid triangle, and `public static double area(double side1, double side2, double side3)` that returns the area using Heron's formula. Write a test program that reads three sides and displays the area if valid.
14. Write a method: `public static int countLetters(String s)` that counts and returns the number of letters in a string. Write a test program that prompts the user to enter a string and displays the letter count.
15. Write a method: `public static int getNumber(char uppercaseLetter)` that returns the phone keypad number corresponding to an uppercase letter. Write a test program that prompts the user to enter a phone number string that may contain letters, translates all letters to digits, and displays the result.
16. TDD Task: Write a method `public static double celsiusToFahrenheit(double celsius)` that converts Celsius to Fahrenheit. Then write a test method that verifies: `celsiusToFahrenheit(0)` returns 32.0, `celsiusToFahrenheit(100)` returns 212.0, and `celsiusToFahrenheit(-40)` returns -40.0.
17. TDD Task: Write a method `public static boolean isEven(int n)` that returns true if n is even. Write a test method that verifies: `isEven(4)` returns true, `isEven(7)` returns false, `isEven(0)` returns true, and `isEven(-2)` returns true.
18. TDD Task: Write a method `public static int factorial(int n)` that returns n! for  $n \geq 0$ . Write a test method that verifies: `factorial(0)` returns 1, `factorial(1)` returns 1, `factorial(5)` returns 120, and `factorial(6)` returns 720.
19. TDD Task: Write a method `public static double circleArea(double radius)` that returns the area of a circle. Write a test method that verifies the result is within 0.001 of the expected value for: `radius=1.0` (expected ~3.14159), `radius=5.0` (expected ~78.5398), and `radius=0.0` (expected 0.0).
20. TDD Task: Write a method `public static String greet(String name)` that returns 'Hello, [name]!'. Write a test method that verifies: `greet('Alice')` returns 'Hello, Alice!', `greet('World')` returns 'Hello, World!', and `greet('')` returns 'Hello, !'.
21. TDD Task: Write a method `public static int max(int a, int b, int c)` that returns the largest of three integers. Write a test method that verifies: `max(1,2,3)` returns 3, `max(3,2,1)` returns 3, `max(2,2,2)` returns 2, and `max(-1,-2,-3)` returns -1.
22. TDD Task: Write a method `public static boolean isPrime(int n)` that returns true if n is prime. Write a test method that verifies: `isPrime(2)` returns true, `isPrime(1)` returns false, `isPrime(17)` returns true, `isPrime(4)` returns false, and `isPrime(0)` returns false.



23. TDD Task: Write a method `public static int sumOfDigits(int n)` that returns the sum of the digits of `n`. Write a test method that verifies: `sumOfDigits(123)` returns 6, `sumOfDigits(0)` returns 0, `sumOfDigits(999)` returns 27, and `sumOfDigits(1000)` returns 1.
24. Write a method: `public static double area(double side)` that returns the area of a pentagon using:  $\text{Area} = (5 \times s^2) / (4 \times \tan(\pi/5))$ . Write a main method that prompts the user to enter the side length and displays the area.
25. Write a method: `public static double area(int n, double side)` that returns the area of a regular polygon with `n` sides of the given length using:  $\text{Area} = (n \times s^2) / (4 \times \tan(\pi/n))$ . Write a main method that prompts the user to enter the number of sides and the side length, then displays the area.
26. Write a method: `public static String format(int number, int width)` that returns a string for the number padded with leading zeros to the given width (e.g., `format(34, 4)` returns '0034'). Write a test program that prompts the user for a number and width, and displays the formatted string.
27. Write a method: `public static boolean leftOfTheLine(double x0, double y0, double x1, double y1, double x2, double y2)` that returns true if point `p2(x2,y2)` is on the left side of the directed line from `p0(x0,y0)` to `p1(x1,y1)`. Also write: `onTheSameLine` and `onTheLineSegment` methods. Write a test program that prompts the user for the three points and displays the position of `p2`.
28. Write a method: `public static double sqrt(long n)` that approximates the square root of `n` using the Babylonian method: `nextGuess = (lastGuess + n / lastGuess) / 2`, starting with `lastGuess = 1`. Stop when the difference between `nextGuess` and `lastGuess` is less than 0.0001. Write a test program that verifies your result against `Math.sqrt()` for several inputs.
29. Write a method: `public static void generateRandomCharacters()` that prints 200 random uppercase letters followed by 200 random single digits, printing 10 characters per line. Write a test program that calls this method.
30. Write a method with the header: `public static double sqrt(long n)` that uses the Babylonian method to approximate square roots. TDD Task: Write a test method that verifies your `sqrt` method returns values within 0.001 of `Math.sqrt()` for inputs: 4, 9, 16, 25, 100, and 144.

**Hint: Write a test program means write a main method**

**Write a test method means write a test file**

# Have fun guys!